Cluster Robust Variance Estimation

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- To conduct hypothesis tests we need the asymptotic distribution of our estimators
- Usal variance estimator is not consistent in case of clustered data
- Cluster robust estimator exist but is not actually robust
- When to use it?

Which is the problem?

- Inside a cluster the scores of 2 individuals are not independent
- Usual White Variance estimator is not consistent

$$\frac{1}{\sqrt{N}}\sum_{i=1}^{N} X_i u_i \xrightarrow{d} \mathcal{N}(0,\ \Omega_N)$$

$$\Omega_N = \underbrace{\frac{1}{N}\sum_{i=1}^{N}\texttt{Var}(X_i u_i)}_{\text{diagonal terms}} + \underbrace{\frac{2}{N}\sum_{i=1}^{N}\sum_{j\neq i}^{N}\texttt{Cov}(X_i u_i, X_j u_j)}_{\text{off-diagonal terms}}$$

Literature Review

- Cameron and Miller (2015); Wooldridge (2003):
  Clustering corrects OLS standard errors for within-cluster dependence; validity relies on **large-G asymptotics**.

- **Ibragimov and Müller (2010)**:
  With fixed or small G, inference based on cluster-level estimates and a t_{G-1} test remains valid under heteroskedasticity.

- **Abadie et al. (2023)**:
  Clustering is justified by **sampling or assignment design**, not residual correlation per se.

- **Key implication**:
  If neither sampling nor treatment is clustered, **do not cluster**; CRVE is often **conservative**, especially with heterogeneous effects.

Representation of Clustered Data

- Sample with N individuals grouped in G clusters:
  $X_{ig} = (X_{11}, \dots, X_{N_1 1},\ X_{12}, \dots, X_{N_2 2}, \dots, X_{N_G G})$
- Each cluster can be seen as a **subpopulation**:

$$X_g = \{X_i : i \in g\}$$

- Individuals inside a cluster share unobserved correlations

- True model:

$$Y_{ig} = X_{ig}\beta_g + u_i$$

- Estimated model:

$$Y_{ig} = X_{ig}\beta_{\text{APE}} + e_{ig}$$

- Residual:

$$e_{ig} = X_{ig}(\beta_g - \beta_{\text{APE}}) + u_i$$

- Hence, **heterogeneous** β_g **generates clustered residuals**.

Assumptions on the DGP

- Two-stage sampling:
  1. sample clusters
  2. sample individuals within clusters
- **Fixed number of clusters (**G **fixed)**
  - clusters are drawn once
  - repeated samples come only from re-sampling individuals
  - **only unit-level randomness**
- **Increasing number of clusters (**$G \to \infty$**)**
  - clusters and individuals are re-sampled each time
  - **two sources of randomness**:

Variance of Cluster-Level Effects

- Assume a population of cluster-specific effects:

$$\beta_g \sim \text{ i.i.d. with } E[\beta_g] = \beta_{\text{APE}}, \quad \texttt{Var}(\beta_g) = \sigma_\beta^2.$$

- We draw G coefficients:

$$\beta = (\beta_1, \dots, \beta_G)'.$$

- Then

$$E[\beta] = \begin{bmatrix} \beta_{\text{APE}} \\ \vdots \\ \beta_{\text{APE}} \end{bmatrix}, \quad \texttt{Cov}(\beta) = \sigma_\beta^2 I_G.$$

Asymptotics in the Number of Clusters

- Sample mean of the cluster effects:

$$\bar{\beta} = \frac{1}{G}\sum_{g=1}^{G}\beta_g \xrightarrow{p} \beta_{\text{APE}}$$

- Central Limit Theorem:

$$\frac{1}{\sqrt{G}}\sum_{g=1}^{G}(\beta_g - \beta_{\text{APE}}) \xrightarrow{d} \mathcal{N}(0, \sigma_\beta^2).$$

- If all β_g were observed, we could estimate σ_β^2 via:

$$\hat{\sigma}_\beta^2 = \frac{G}{G-1}\sum_{g=1}^{G}(\beta_g - \bar{\beta})^2.$$

From True β_g to Estimated $\hat{\beta}_g$

- In practice, β_g is not observed; we estimate it from data.
- For each cluster g:

$$\sqrt{N_g}\,(\hat{\beta}_g - \beta_g) \xrightarrow{d} \mathcal{N}(0, V_g),$$

  where

$$V_g = (X_g' X_g)^{-1} E_g[X_g' u_g u_g' X_g](X_g' X_g)^{-1}.$$

- Across clusters we thus have G asymptotically normal estimators, each centered at its own β_g with cluster-specific variance V_g.

Interpretation: Signal + Noise

- Write each estimator as
$$\hat{\beta}_g = \beta_g + e_g,$$
with
$$\sqrt{N_g}\, e_g \xrightarrow{d} \mathcal{N}(0, V_g), \quad E[e_g] = 0.$$
- The vector $\hat{\beta}$ can be viewed as **noisy observations** of the underlying cluster parameters, with noise coming from within-cluster sampling.

Decomposition Around β_{APE}

2 Montecarlo Experiment

- Model the true effects as

$$\beta_g = \beta_{\text{APE}} + d_g, \quad d_g \sim \text{i.i.d. } \mathcal{N}(0, \sigma_\beta^2).$$

- Then

$$\hat{\beta}_g = \beta_{\text{APE}} + d_g + e_g.$$

- The cluster mean equals the pooled OLS estimator:

$$\hat{\beta}_{\text{pols}} = \frac{1}{G}\sum_{g=1}^{G}\hat{\beta}_g = \beta_{\text{APE}} + \frac{1}{G}\sum_{g=1}^{G} d_g + \frac{1}{G}\sum_{g=1}^{G} e_g.$$

- As $G \to \infty$, both averages of d_g and e_g converge to zero in probability.

Monte Carlo Decomposition of $\mathrm{Var}(\hat{\beta}_{\text{ols}})$

- Goal: assess how well

$$\mathrm{Var}(\hat{\beta}_{\text{ols}}) \approx \mathrm{Var}(\bar{d}_g) + \mathbb{E}\left[\mathrm{Var}(e_g)\right]$$

  holds in finite samples.
- Two components:
  - $d_g = \beta_g - \beta_{\text{APE}}$ (heterogeneity)
  - $e_g = \hat{\beta}_g - \beta_g$ (sampling error)

Simulation Design

2 Montecarlo Experiment

- Loop over different **numbers of clusters** G and **cluster sizes** N_g.
- For each (G, N_g), run **1000 Monte Carlo replications**.
- For each replication:
  1. Draw heterogeneous coefficients β_g.
  2. Generate a population with G clusters.
  3. Sample N_g individuals from each cluster.
  4. Estimate all $\hat{\beta}_g$ and pooled $\hat{\beta}_{\text{ols}}$.

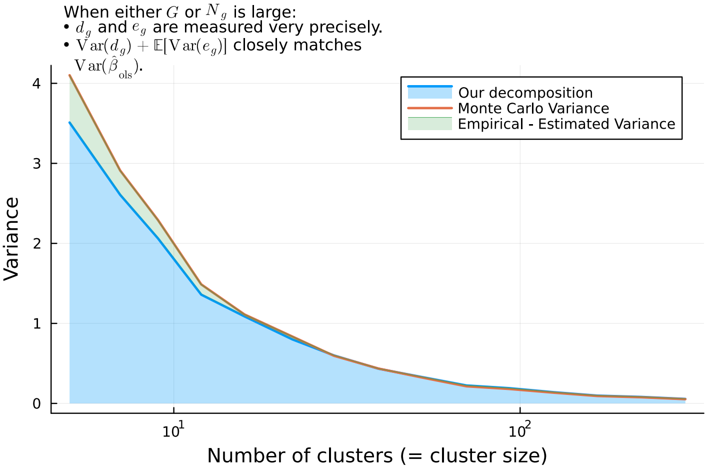
Recorded Quantities

2 Montecarlo Experiment

- For every repetition and design (N_g, G):
  - $e_g = \hat{\beta}_g - \beta_g$ (estimation errors)
  - $d_g = \beta_g - \beta_{\text{APE}}$ (heterogeneity terms)
  - $\hat{\beta}_{\text{ols}}$ (pooled estimator)
  - White and CRVE variance estimates of $\hat{\beta}_{\text{ols}}$.
- For each (N_g, G) we compute:
  - $\mathtt{Var}(\hat{\beta}_{\text{ols}})$ (empirical variance)
  - $\mathtt{Var}(d_g)$ and $\mathbb{E}[\mathtt{Var}(e_g)]$
  - relative discrepancy.

Performance of the Decomposition

2 Montecarlo Experiment



CRVE–White Differences as Heterogeneity Measure

- We test whether the **difference** between CRVE and White variance estimators captures the heterogeneity component

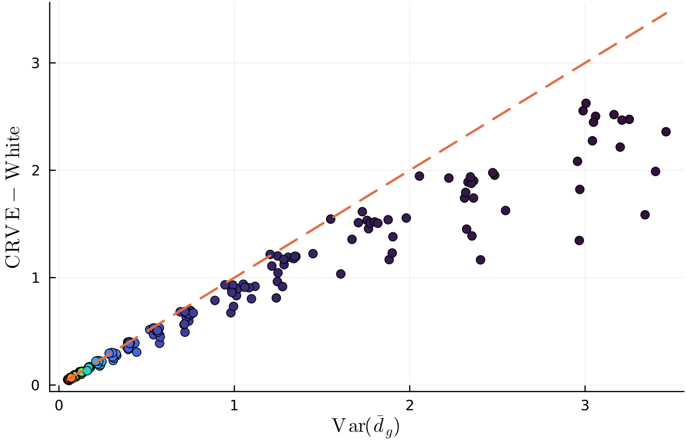
$$\mathrm{Var}(\bar{d}_g), \quad d_g = \beta_g - \beta_{\text{APE}}.$$

- For each design (N_g, G):
  - compute average CRVE variance,
  - compute average White variance,
  - take their difference,
  - compare it to $\mathrm{Var}(\bar{d}_g)$ from the simulations.

Simulation-Based Comparison

- For every (N_g, G):
  - horizontal axis: $\mathrm{Var}(\bar{d}_g)$ (variance of mean deviations),
  - vertical axis:
$$\mathbb{E}[\widehat{\mathrm{Var}}_{\text{CRVE}}] - \mathbb{E}[\widehat{\mathrm{Var}}_{\text{White}}].$$
- Each point is a design (N_g, G):
  - color intensity encodes G (lighter = larger G).
- The 45-degree dashed line is the **benchmark** of perfect equality.

CRVE − White vs Var(mean(d_g))



- Points lie very close to the 45-degree line:
  - CRVE–White differences almost exactly recover $\mathrm{Var}(\bar{d}_g)$.
- Overall, the simulations support:

$$\mathbb{E}\left[\widehat{\mathrm{Var}}_{\text{CRVE}}\right] - \mathbb{E}\left[\widehat{\mathrm{Var}}_{\text{White}}\right] \approx \mathrm{Var}(\bar{d}_g),$$

and this relationship strengthens as the number of clusters G grows.

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Conclusions

- The Variance of OLS can be decomposed as:

$$\mathtt{Var}(\hat{\beta}_{\text{ols}}) \approx \mathtt{Var}(\bar{d}_g) + \mathbb{E}\left[\mathtt{Var}(e_g)\right]$$

- CRVE estimates White + $\mathtt{Var}(\bar{d}_g)$
- So if $\mathtt{Var}(\bar{d}_g) \approx 0$ we should use White
- This happens if:
  - There is no resampling (G is fixed)
  - G is really high

Thanks!

Abadie, Alberto, Susan Athey, Guido W. Imbens, and Jeffrey M. Wooldridge. 2023. "When Should You Adjust Standard Errors for Clustering?" *The Quarterly Journal of Economics* 138 (1): 1–35. https://doi.org/10.1093/qje/qjac038.

Cameron, A. Colin, and Douglas L. Miller. 2015. "A Practitioner's Guide to Cluster-Robust Inference." *Journal of Human Resources* 50 (2): 317–72. https://doi.org/10.3368/jhr.50.2.317.

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